Climate inequality

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Abstract

This paper characterizes the relationship between anthropogenic climate change and economic inequality within countries, a subject which intersects two of the defining social challenges of the 21st century but remains remarkably understudied. To address this urgent research gap, I build on cutting-edge methods for estimating climate impacts and apply these refinements to newly available distributional national accounts data. I document new evidence that temperature shocks exacerbate income inequality within countries, an effect driven by disproportionate reductions in income among the poorest segments of already-poor countries. To better understand the implications of this incidence pattern for the welfare economics of climate change, I then integrate these results over observed income distributions and use output from the newest generation of global climate models to perform a counterfactual analysis estimating anthropogenic contributions to these climate inequalities. These findings represent the most comprehensive characterization yet of the economic regressivity of climate change within countries and globally.

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1. Introduction

Mitigating anthropogenic climate change and reducing economic inequality are widely recognized as two of the most pressing social issues facing the world in the 21st century. They are foundational to the 2030 Agenda for Sustainable Development adopted by all United Nations members, whose Sustainable Development Goals explicitly resolve to "take urgent action to combat climate change and its impacts" and "reduce inequality within and among countries" (United Nations, 2015). These are also understood to be intimately related and intersecting objectives: the Intergovernmental Panel on Climate Change recognizes "addressing challenges and widening opportunities between and within countries" (Masson-Delmotte et al., eds, 2018) as necessary in any sufficient policy response to climate change while the World Inequality Report remarks that "climate mitigation is largely a distributional issue, not only between countries but also within them" (Chancel et al., 2022).

Despite this appreciation of the linkages between the two vitally important phenomena, there is remarkably little economic literature which considers them jointly. The vast and influential body of work on climate impacts which has emerged in recent decades has largely followed a cost-benefit framework which evaluates the climate change externality on the basis of highly aggregated net impacts, typically at a global, national, or sectoral level. Even among the subset of entries which depart from this distributional agnosticism, the scope of analysis has been characterized by a steep tradeoff between geographic disaggregation and breadth. For example, Hsiang et al. (2017) is a particularly impressive empirical undertaking which integrates fine-scaled agricultural, crime, energy, mortality, and labor data to determine that climate risks in the continental United States are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts. But for more generalizable claims about the inequality implications of global warming, analyses must extend to settings which lack comparably rich or granular data. Representing the opposite end of this tradeoff, Diffenbaugh and Burke (2019) report that the ratio between the top and bottom deciles of the global income distribution is 25% larger than it would be had anthropogenic contributions to climate change been constraiend to 1960 levels, all else held equal. However, this measure is constructed by assigning all individuals their country's per capita GDP, a necessary practice given the constraints of a dataset with only 160 differentiated units of vastly different population sizes, but one which nonetheless imposes a very conservative lower bound on total inequality (Cowell, 2011).

In fact, Chancel et al. (2022) reports that between-country inequality has

comprised a rapidly diminishing share of global inequality since the Asian and African decolonization movements of the second half the 20th century, sharply reversing a trend which had persisted since the onset of the industrial revolution. By the decomposable Theil measure, between-country differences now constitute only 30% of global income inequality with the rest explained by within-country disparities. In this light, a characterization of global 'climate inequality' which restricts attention to its cruder between-country components not only constrains attention to the aggregate production of countries rather than the welfare of peoples, but also becomes increasingly irrelevant to the global inequality that will characterize the 21st century should current trends persist. A richer understanding of these within-country disparities may also improve prospects for an effective response to the impacts of climate change since democratic pressures and a richer portfolio of unilateral policy instruments are more effectively mobilized within a sovereign polity than across a Westphalian international political economy (Barrett, 2003; Levin et al., 2009; Keohane and Victor, 2011).

This paper attempts to fill this critical research gap by providing the most comprehensive characterization yet of the inequality implications of climate change, placing particular emphasis on its within-country features which remain relatively unexplored in existing work. To do so requires overcoming several challenges which have contributed to the sparsity of entries in this urgent but inactive area of study. First, prevailing empirical methods commonly used to estimate climate impacts require refinement. Even in standard country-level analyses using the same sources of data, estimated impacts of climate change are known to diverge by orders of magnitude depending on how the modeler chooses to specify the "damage function" causally relating variation in weather to economic outcomes of interest (Pindyck, 2013). To avoid transmitting this imprecision to our new research setting, I first build upon cutting-edge impact models to address outstanding methodological shortcomings flagged in recent literature. I also draw from recent empirical macroeconomic literature to derive a direct statistical test for the persistence dynamics whose unfalsifiability has largely driven the extreme sensitivity of cumulative impacts reported in the literature (Newell et al., 2021). This methodological contribution is presented in Section Section 2 through an expositional application to standard aggregate impacts and may be of general interest to the climate impacts research community.

To make progress on the disaggregation-breadth tradeoff described above, I apply this refined apparatus to recently published data from the World Inequality Database which systematically harmonizes household survey, national accounts, and tax return data to provide comparable percentile-level

estimates of income for a wide set of countries. I document new empirical evidence in Section 2 that demonstrate that within-country inequality effects of identified temperature shocks are significant and driven primarily by disproportionately negative and relatively persistent impacts to lower income quantiles in already-poor countries. Interestingly, an asymmetrical hump-shaped incidence pattern is recovered which implies a net transfer of national income shares towards the middle of the income distribution. This suggests a temperature incidence pattern that is regressive in the sense of transferring income shares away from bottom towards the middle but that also measures a surprising degree of exposure of top incomes to environmental shocks relative to the middle classes. Overall, the empirical analysis suggests a significant redistribution shares of near-certainly reduced total income.

In Section 4 then integrate these reduced-form results with observed distributions of income and the outputs of state-of-the-art climate models to estimate contributions of the incidence of temperature shocks to the global distribution of income since 1980, the first year our inequality data is available. When impacts are aggregated globally, are unambiguously negative overall in the sense of almost uniformly implying absolute losses of income across the income distribution.

and to decompose these effects into anthropogenic and natural causes. In Section 4.2, I describe the simulation data and methods used to perform this counterfactual exercise, which draws from the relatively new subfield of attribution climate science. I report the results of this analysis in Section 4.3, finding that anthropogenic factors have almost certainly lowered incomes across the entire global income distribution since 1980 with relative net transfers away from the bottom quantiles driven by relativley large climate impacts on within-country inequality compared to between-country inequality. Section 5 concludes by considering how these new findings may inform the design of climate policy and inspire new lines of climate economic research recognizing distributional impacts as first-order concerns complementary to the prevailing cost-benefit paradigm.

1.1 Related literature

There is a vast empirical literature estimating climate "damage functions" which causally relate variation in weather variables such as temperature to changes in aggregate economic outcomes. Those which employ a "top-down" empirical approach[1] typically measure outcomes using widely available data

[1]Top-down approaches regard an aggregate outcome as a representative measure of the relevant social impact of climate, remaining agnostic about non-market impacts. This contrasts

on aggregate production enabling an analysis of heterogeneous impacts at the country level. While a lack of consensus on estimated magnitudes is well-documented, country-level damage functions typically feature similar non-linearities in temperature wherein marginal damages from global warming increase with a country's baseline temperature, implying a concentration of damage and risk in already-poorer countries whose populations reside closer to the tropics. A subset of these studies specifically study the implications of this between-country dimension of inequality directly, such as Diffenbaugh and Burke (2019) which uses the damage function introduced in Burke et al. (2015) to measure the degree to which global warming observed since 1960 has exacerbated between-country inequality over the same period. Other work highlights the inverse relationship between contributions to cumulative global greenhouse gas emissions and their exposure to their social consequences (Chancel and Piketty, 2015; Ricke et al., 2018; Kotz et al., 2024).

Similar non-linearities emerge at the micro level across a range of outcomes as varied as labor supply (Graff Zivin and Neidell, 2014), agricultural yields (Schlenker and Roberts, 2009), standardized test performance (Goodman et al., 2020), and even social media profanity (Baylis et al., 2018). Carleton and Hsiang (2016) overviews the proliferation of multidisciplinary studies in this category whose results are generally suggestive of patterns of increased inequality within countries similar to those observed in the aggregate.

Nonetheless, the literature on climate impacts on economic inequality *within* countries remains comparatively sparse in large part due to an absence of adequate research infrastructure. Estimation of distributional equivalents to the aggregate damage functions requires data comparable in credibility and coverage to domestic production data whose construction across countries follows rigorous standards set forth by the international System of National Accounts (European Commission et al., 2008). Historically, distributional measurement of welfare has lacked and arguably been crowded out by the wider interest and investment in measuring aggregate production (Jorgenson, 2018). While secondary inequality data products compiling country-year inequality variables from a wide array of official and survey sources date at least as far back as Deininger and Squire (1996), the validity of these products for cross-country causal analysis have been long-known to be limited by inconsistencies across primary sources in production and availability Atkinson and Brandolini (2001). For example, the most recent rigorous evaluation I could find of these secondary datasets concluded that results derived from the Standard-

with "bottom-up" enumerative approaches in which impacts to a "subsector" such as mortality, energy consumption, or agriculture are quantified and possibly integrated with other sectoral results.

ized World Income Inequality Database (Solt, 2020) should be regarded as "not sufficiently credible" and that analyses using the United Nations' long-running World Income Inequality Database (UNU-WIDER, 2021) should at best be accompanied by the practitioner's justifications for their sample and variable selection criteria (Jenkins, 2015).

A handful of works have managed to contribute to the study of within-country inequality impacts by using idiosyncratic data available for individual countries. A particularly impressive example is the integration of agricultural, crime, energy, mortality and labor data in Hsiang et al. (2017) to map the geography of climate risks in the continental United States. The authors find that risks are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts. Marx (2024) finds that years with more days above 30 are associated with wider disparities between richer and poorer cantons (an administrative unit described as equivalent to roughly 10 municipalities). Dasgupta et al. (2023) uses data from perhaps the most unequal country in the world to project a 3-6 Gini point increase in South Africa by 2100 under a moderate warming scenario.

The research programs of Thomas Piketty, Emmanuel Saez, and their co-authors since the turn of the century are widely credited with revitalizing public and academic interest in the measurement of within-country inequality. Beginning with work tracing the long-run evolution of income and wealth inequality in France (Piketty, 2003) and the United States (Piketty and Saez, 2003), their mobilization of extensive tax returns data has enabled credible measurement of top incomes over long time horizons, stimulating new programs of research and equipping the general public with new political language around which to mobilize (Jones, 2015). This paper takes advantage of the cross-country research apparatus that has rapidly developed and been made accessible in recent years with the intention of maturing economic analysis of climate change in a similar manner. Palagi et al. (2022) is the only other climate-economic study I could locate which makes use of the same "distributional national accounts" (World Inequality Lab, 2024) data used in this study. The authors find that extreme precipitation negatively impacts low-income individuals much more severely in countries more dependent on agriculture. I elaborate on the construction and shortcomings of this dataset in Section 2.1.

I locate this paper in the context of a growing demand for distributional considerations in economic analysis of climate change and the design of climate policy. For example, Chancel et al. (2023) estimates that within-country inequality of carbon emissions have recently exceeded the equivalent "carbon inequality" across countries and advocates for targeted carbon taxation programs based on individual footprints to more efficiently internalize the

climate change externality. The modern SCC framework accommodates equity considerations by concavifying the underlying utility function so that the welfare weights assigned to individuals are inversely related to their relative level of consumption; a logarithmic specification is commonly chosen, for example. Variations of this progressive utilitarianism have been institutionalized to some degree in a handful of countries including recently in the United States (Office of Management and Budget, 2023). Theoretically, the way equity weighting is implemented should result in substantially higher SCC estimates depending on the extent to which the distributive effect of climate change is modeled as regressive[2]. This paper provides the first credible estimates of this regressivity at a near-global level and at subnational resolution.

2. Empirical analysis

2.1 Data

Inequality

Our inequality measures derive from the grouped income data reported by the World Inequality Database World Inequality Lab (2024), the successor to the tax data-based World Top Income Database which launched in 2011 with data on over 30 mostly Western countries. A critical development since has been the development of the system of "distributional national accounts" (DINA) (Piketty et al., 2018) for standardizing the integration of tax, national accounts, and survey data to more fully characterize income distributions Piketty et al. (2019). This newer generation of inequality data extends coverage within those countries—now claiming to capture "100% of income" in the United States (Piketty et al., 2018)—and rapidly extending coverage and comparability to countries with relatively limited available data. By 2019, historical data for over 120 countries had been incorporated into the database. As of this writing, their equal-split adult pre-tax income series was available for 215 countries and territories collectively representing almost the entire global population with distributional data imputed every year since 1980.

From this income data, we construct country-year panels of summary inequality measures such as the Gini coefficient, mean logarithmic deviation, and the Theil index. Appendix A1 describes the construction of the latter two measures in more detail.

[2]Prest et al. (2024) show that one proposed implementation would increase the US government's official SCC estimate by a factor of eight

Notably for the percentile-level panels, the income data exhibits substantial bottom-coding with almost every country-year reporting exactly zero income for at least the bottom five percentiles of the pre-tax income distribution. This reflects adult unemployment but also does not account for wages earned in the informal sector and post-tax transfers, savings, and communal safety nets. It also positively biases estimated changes in welfare among the poor since zero incomes mechanically cannot decrease. For these reasons among others, much less-available measures of consumption expenditures are considered much preferred proxies for the welfare of individuals near the bottom of income distributions, particularly in developing countries (Meyer and Sullivan, 2003).

Given these inherent disadvantages of income measures for representing poverty, we intentionally do not make specific claims about the poverty implications of our findings even though this is obviously a first-order distributional concern. We aggregate the incomes of the bottom 10 percentiles in all analyses to consider the bottom 10% of the income distributions a single grouping and consider estimated impacts for this quantile to be a conservative bound on true impacts on the very poorest. The accompanying metadata quantifies the availability and reliability of each category of input source by country, indicating that the input mix of these sources varies substantially by country. All incomes are denominated in 2023 US dollars using the provided PPP-adjusted exchange rates.

Weather

We merge all our panels with a panel of annual population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). This data corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Department of Civil and Environmental Engineering/Princeton University, 2006). The GMFD reports historical weather data for the entire global surface at 0.25-degree resolution[3] for every three hours from the start of January 1, 1948 to the end of December 31, 2015.

2.2 Methods

This study employs an empirical strategy which builds on recent innovations in the estimation of reduced-form climate damage functions by incorporating

[3] At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

augmentations based on recent empirical macroeconomics literature on the estimation of impulse responses.

Identifying temperature shocks

It has been customary in the empirical climate impacts literature to use polynomial functions of weather variables such as temperature or rainfall as primary explanatory variable of interest. These constructions are motivated by the intuition that annual fluctuations in temperature are plausibly exogenous with respect to most economic outcomes except through an indirect channel where expanded production correlates with weather through its association with greater emissions of greenhouse gases.

Bilal and Känzig (2024) and Nath et al. (2024) define shocks as the residual from autoregressions of temperature, which they motivate out of concern that estimation using levels of temperature may be confounded by their serial correlation or through its causal relation to their outcome variable (GDP) which correlates with emissions which in turn induce local warming.

Ramey (2016) outlines conditions for candidate treatment variables to qualify as shocks which can be used for valid dynamic causal inference. In short, they must correspond to unanticipated movements in exogenous variables, be exogenous with respect to current and lagged endogenous variables, and be uncorrelated with other exogenous shocks included in the regression model. We follow Kahn et al. (2021) in defining shocks $\widehat{\tau}_{it}$ with respect to an M-period moving average of local temperature:

$$\overline{T}_{it} := \frac{1}{M} \sum_{m=1}^{M} T_{i,t-m} \tag{1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{1b}$$

Shocks defined this way satisfy Ramey's exogeneity conditions when lags of the shock are included as controls in the model being estimated. An alternative used in recent years is the construction of Bilal and Känzig (2024) and Nath et al. (2024) which define shocks as the residual from autoregressions of temperature, which they motivate out of concern that estimation using levels of temperature may be confounded by their serial correlation or through its causal relation to their outcome variable (GDP) which correlates with emissions which in turn induce local warming.

I consider this simple construction to be the most compelling formulation of a temperature shock for several reasons. First, for sufficiently large M, it

has an appealing coherence with the climatological convention of using long-run moving averages of weather realizations to anchor the climate at a fixed point in time. Most commonly, these "climate normals" are defined as 30-year averages to account for the fact that even a stationary climate may exhibit substantial medium-term fluctuations in weather. The most economically relevant example of this natural variability is the El Niño Southern Oscillation, an ocean-warming phenomenon which can vary average global surface temperatures by as much as 0.4 transitioning between its three phases in cycles which range from 2-7 years. Other oscillations can span multiple decades. Normals are used to distinguish natural weather variability from systematic climate change.

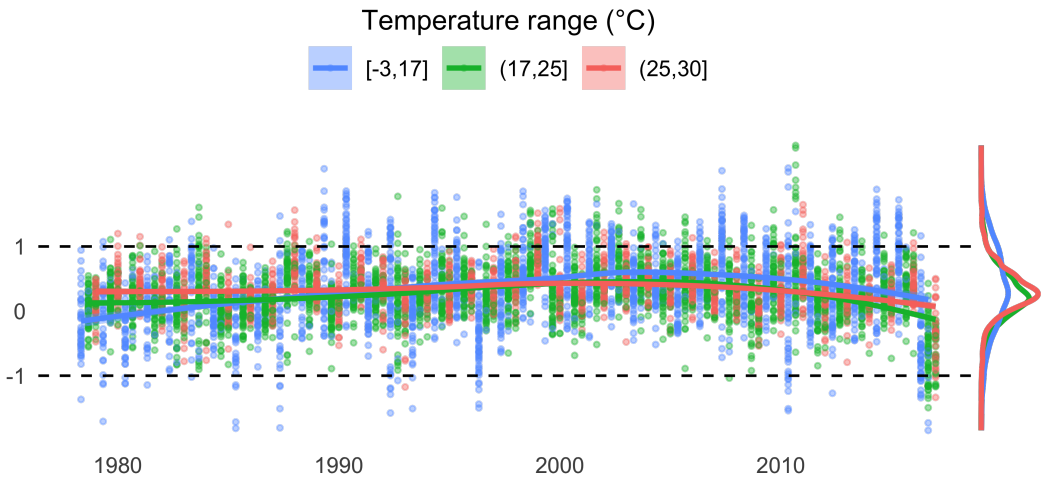
We interpret this state variable $\overline{T}_{it}$ as a summary statistic parametrizing the unobserved probability distribution from which the weather realized in country i at time t is drawn, which may refer to as the local "climate" which may itself vary over time. Systematic trends in the state variable represent "climate change" as experienced by country i. In the historical simulation exercise to follow, we consider decompositions of observed changes in climate into natural and anthropogenic components.

To the extent that local expectations about the climate are anchored by recent experience, shocks defined this way may be interpreted as deviations relative to adaptive expectations; a given temperature level may register as a "hot" shock one year but a "cold" shock in the same country decades later if the intermediate period is characterized by systematically warmer weather that its inhabitants have internalized either through some concept of informational updating or since it constitutes the entire life experience of a younger generation. I discuss other appealing features of this construction in the next subsection on estimation of temperature effects.

Since the moving average is centered on the period $t - \frac{M}{2}$, the state variable technically corresponds to a lag of the true state; we cannot center the moving average on period t since it would include future observations for any $M > 0$ in a manner similar to the argument against use of Hodrick-Prescott trend filters in Hamilton (2018). Over the span of our historical data, we find that the median country observes, on average, a 0.26°C (IQR: 0.18-0.36) increase in the 30-year moving average every 15 years, a non-trivial amount. We prefer to account for this by including lagged shocks but there is also a potentially attractive notion of informational inattention in retaining this imperfect adaptation. A compromise between the two can be achieved by reducing M or imposing a weighted average scheme so that economic agents are more responsive to more recent observations.

Figure 1 depicts country time series of the resulting temperature shocks

Figure 1: **Country-level time series of identified temperature shocks**



Points correspond to shocks observed in individual country-years. The three colors correspond to an equal-sized binning of country-years by their moving-average temperature. Flexible local regression curves describe time trends split by this binning. Marginal distribution plots on the right show that colder country-years observe greater variance in shocks.

with colors and flexible local regression curves corresponding to a splitting of all country-years into three groups defined by their average temperature. The positive values of the local regression curves over almost the entire observation period capture the idea that systematic warming is characterized by relatively frequent "hot" shocks relative to "cold" even if expectations about weather adapt to recent weather. Intervals characterized by smaller shocks correspond to relative stability in the climate. The marginal distribution plots on the right demonstrate that colder seasonal countries experience much larger temperature shocks than their warm tropical counterparts, a reflection of both greater natural weather variability and the fact that anthropogenic warming is generally greater in colder regions furthest from the equator.

Estimating responses to a temperature shock

We adapt the semi-parametric local projections method pioneered by Jordà (2005) to estimate state-dependent impulse responses to a temperature shock. As Jordà (2023) notes, the method has found growing appeal outside its home field of applied macroeconomics as its flexibility (Cloyne et al., 2023), relevance to the potential outcomes framework (Dube et al., 2024), and robustness to misspecification and non-stationarity (Montiel Olea and Plagborg-Møller, 2021; Montiel Olea et al., 2024; Piger and Stockwell, 2023) have been established in recent econometric literature. Berg et al. (2024), Bilal and Känzig (2024), and Nath et al. (2024) are other climate papers which apply the method to estimate dynamic economic responses to temperature shocks.

These features are particularly useful for addressing several outstanding identification concerns flagged in reviews of impact estimation methods. We use an application to country-level GDP impacts to aid exposition of the methods used in this study which will also serves to familiarize the reader with the literature on aggregate impacts to contextualize the new within-country impacts which constitute the main contribution of this paper.

We use panel data to estimate dynamic responses to temperature for several distributional outcomes. For now, we take y_{it} to represent the logarithm of GDP for country i in year t. The local projections method entails estimating $H+1$ iterations of the following single-equation model corresponding to different projection horizons $h \in \{0, 1, ..., H\}$:

$$\begin{aligned} \Delta_h y_{i,t+h} &:= y_{i,t+h} - y_{i,t-1} \\ &= \beta_{1h}\widehat{\tau}_{it} + \beta_{2h}\widehat{\tau}_{it} \cdot \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + u_{i,t+h} \end{aligned} \tag{2}$$

Each projection horizon h thus attributes a main effect β_{1h} representing the impact of a unit pulse of temperature on the "long difference" of y projected h periods after the shock. An interaction term allows the effect to scale linearly with the country-year's long-run average temperature $\overline{T}_{it}$. This state-dependency is motivated by the standard result in the climate impacts literature that the economic effect of a temperature shock is increasingly negative for warmer climates and we expect this coefficient to be negative. Finally, the inclusion of a main effect λ_h associated with the time-varying state variable allows for the possibility that a systematic change in the climate may impact expected growth independent of its impact on the distribution of shocks. For example, if a hot country stabilizes at a climate that is 1°C warmer after an intermediate period of warming, it is intuitive that economic productivity may be systematically lower even when given sufficient time to maximally adapt.

The matrix $\mathbf{Z}_{it}$ includes lagged controls $\{\widehat{\tau}_{i,t-j}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta y_{i,t-j}\}_{j=1}^{p}$. The inclusion of p lags of the shock $\widehat{\tau}$ accounts for potential autocorrelation in the treatment, validating its use as a shock as described in the previous subsection. Interactions of these lagged shocks with the state variable capture potential state-dependencies in these autocorrelations. Lags of the outcome are included as a guard against non-stationarity. Finally, the set μ_i and η_t of standard linear fixed effects are included to absorb time-invariant country-specific effects and common year-specific effects.

Collecting estimates of the coefficients of interest $\{\widehat{\beta}_{1h}, \widehat{\beta}_{2h}\}_{h=0}^{H}$ and adapting the notation of Jordà (2023), local projections imply a family $\mathcal{R}_{\tau \to y}(h; \overline{T}$ of impulse responses as a function of the state variable $\overline{T}$:

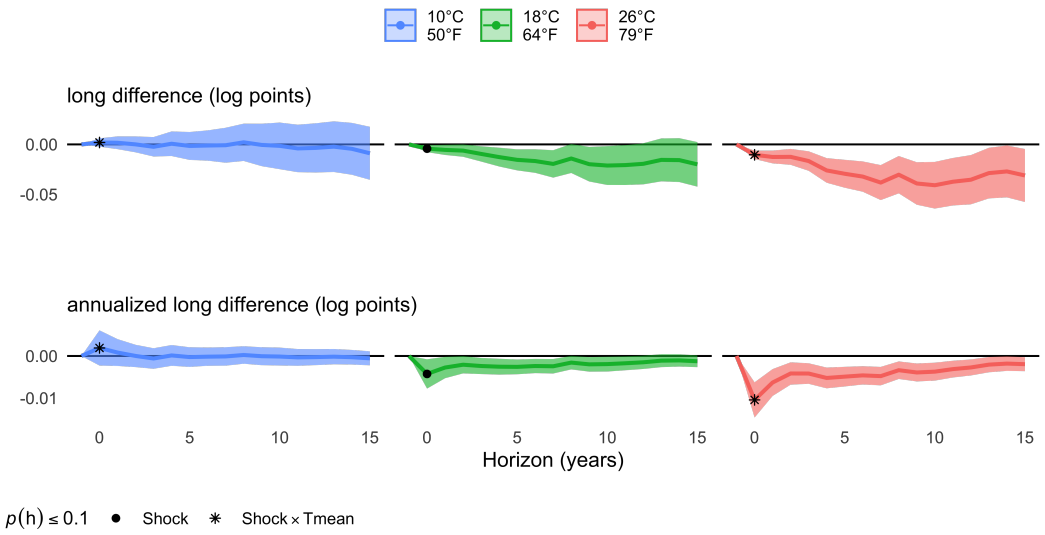
$$\mathcal{R}_{\tau \to y}(h; \overline{T}) := \widehat{\beta}_{1h} + \widehat{\beta}_{2h} \cdot \overline{T} \tag{3}$$

For example, the top row of Figure 2 depicts three different IRFs which each result from the same regressions but are distinguished by different assumed values of the state variable. These representative states correspond to the 20th and 80th percentiles of country-year temperatures in our sample (10 colored blue and 26 colored red) and their average (18, roughly the 40th percentile and colored green). Note that for expositional convenience, we have transformed our state variable to center on the intermediate reference temperature of 18°[4] so that the responses in green correspond to the sequence of estimated main effects $\widehat{\beta}_{1h}$ while the responses in blue and red correspond

[4]To be specific, we redefine the state variable as $\overline{T}_{it} := \frac{1}{30} \sum_{m=1}^{30} (T_{i,t-m} - 18)$ with temperatures expressed in degrees Celsius. Under this transformation, the blue and IRFs correspond to the states $\overline{T}_i = -8$ and $\overline{T}_i \pm 8$ respectively.

incorporate interaction effects corresponding to 8° deviations from this reference state, calculated as $\widehat{\beta}_{1h} \pm 8\widehat{\beta}_{2h}$. These colors and reference temperatures will be used throughout the rest of this paper.

Figure 2: **GDP impulse response to a 1°C shock**



The plots in the top row correspond to estimates of impulse responses of GDP estimated by the $H+1$ local projections regressions specified in (2) for three representative temperatures depicted in different colors. The plots in the bottom row depict the same functions annualized by dividing estimates by $h+1$. All figures in this paper adopt the same color-temperature correspondence where relevant. Points in black correspond to point estimates deemed significant by the joint hypothesis test described in Section 2.2 with filled circles corresponding to significant main effects and asterisks corresponding to significant interaction effects.

It is worth taking account of the restrictions imposed by the model described in (2). The flexibility of the semi-parametric approach is attractive in the context of a climate impacts literature where the magnitude of estimates are known to be largely driven by the specification of the damage function (Newell et al., 2021; Chang et al., 2023). For example, two papers projecting the proportional loss in cumulative global production by 2100 arrive at point estimates of -3.4% (Casey et al., 2023) and -20% (Burke et al., 2015) under a worst-case warming scenario. Almost all this divergence stems from

disagreement over whether to specify the outcome variable in levels (e.g., log GDP) or as growth (e.g., first differences in log GDP). This choice amounts to assuming whether temperature imposes a temporary or permanent effect on the trajectory of GDP. Evaluating the appropriate persistence with which to characterize the temperature effect within fully parametric models requires the addition of sufficient lags of the explanatory variables in order to determine whether cumulative effects are distinguishable from 0. As the horizon being considered increases, the efficiency gains from parametric specification are diminished to accommodate increasingly imprecise dynamic effects. The benefits of lag augmentation are further constrained by non-linear functional forms and autocorrelation in the explanatory variables, both general features of damage functions where absolute temperature is used as an explanatory variable. Indeed, (Burke et al., 2015) conclude that the remaining identifying variation "cannot reject the hypothesis that this effect is a true growth effect nor the hypothesis that it is a temporary level effect."

By comparison, the local projections method imposes no restriction on the evolution of a given IRF over time since the effects associated with each projection horizon are estimated from different regressions and subsamples[5] In addition, the local projection specified in long differences has been shown to be particularly robust to biases associated with persistent and unit root processes Piger and Stockwell (2023); Jordà (2024) known to plague GDP panel regressions specified in levels (Nelson and Plosser, 1982; Campbell and Mankiw, 1987). Altogether, the method is relatively robust to different forms of misspecification while remaining mostly agnostic about the underlying data-generating process. Compared to, for example, a growth regression which infers permanent persistence from effects contemporaneous to the shock, the LP method directly regresses observed $h-$-period long differences in the outcome variable on the treatment variable so that the presence of level and growth effects is inferred directly from observed h-period responses summarized by the IRF $\mathcal{R}_{\tau \to y}(h; \overline{T})$:

1. **Level effect**
    - One-period level effect: $\mathcal{R}_{\tau \to y}(h; \overline{T}) = 0 \ \forall \ h > 0$
    - Persistent level effect: $\exists \ h^* > 0$ such that $\mathcal{R}_{\tau \to y}(h; \overline{T}) = 0 \ \forall \ h > h^*$
2. **Growth effect**

[5]Residuals across the projection horizon are generally autocorrelated but do not affect consistency of point estimates.

- Simple growth effect: $\mathcal{R}_{\tau \to y}(h; \overline{T})$ converges to a non-zero constant
- Divergent growth effect: $\mathcal{R}_{\tau \to y}(h; \overline{T})$ never converges to a constant

The point estimates in the first row of Figure 2 depict a familiar pattern of near-null temperature effects in cold climates, mild effects in mid-temperature climates, and largest effects in hot climates. The evolution of these impacts over time is less commonly depicted and here produce a pattern qualitatively similar to those reported in Nath et al. (2024) wherein per capita GDP appears to stay persistently depressed over the entire projection horizon. The second row depicts the same IRFs scaled down by the corresponding horizon $h + 1$ to more clearly depict convergence behavior relative to the contemporaneous effect of the shock.

Of course, the standard bias-variance tradeoff applies so that the attractive flexibility features described above come at the cost of efficiency; IRFs estimated by local projections are generally noisier than their fully specified counterparts and structural VARs. Different compromises in this tradeoff can be obtained by imposing additional restrictive parametric assumptions such as the functional approximation approach of Barnichon and Matthes (2018) which can substantially narrow the error bands.

Importantly, the LP model imposes a linearity assumption on effects estimated within the same horizon. Since the shock $\widehat{\tau}_{it}$ and the interaction $\widehat{\tau}_{it} \cdot \overline{T}_{it}$ enter additively into (2), both effects are assumed to scale in proportion to the size of the shock so that a 1 shock has exactly twice the effect of a 0.5° shock and the exact negative of the effect of a -1 shock. In reality, non-linearities abound in the interactions between social and climate system (Dietz et al., 2021). Additionally, the linearity assumption imposes the same symmetry in the interaction term so that relative to the reference temperature, estimated interaction effects for states a given number of degrees warmer than the reference temperature will be the negative of those for states the same number of degrees cooler. In Figure 2, this is reflected in the fact that for a given horizon h, the impacts in green will always be the exact average of the impacts in blue and red since they are estimated from the same linear regression model. Both of these assumptions can be relaxed in alternative specifications in which the shock and the interaction enter non-linearly, but we do not consider these alternatives in the current study.

It is worth noting that our empirical estimates are expressed in terms of responses to a 1°C shock, which is substantially larger than a shock we would expect to observe in a single year as we showed in Figure 1. The median absolute value of a shock (whether positive or negative) varies by the rate of climate change which will in turn vary across states and time. For the warmest

33% of country-years, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37 in the ensuing decades and so hot-country effects can be understood as corresponding to a shock 3-5 times larger than one would expect to occur in those climates in a given year. For the coldest 33% of country-years, the median magnitude of a shock was about 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as just 2-3 times larger in magnitude than the median shocks.

Global inequality effects of climate change thus are not just dependent on the shape of the impulse response responses across quantiles but also the distribution of warming over space and time as well as the actual distributions of incomes. The simulation exercises to be described in **??** are valuable because they combine four relevant climate inequalities: inequality of temperature effects across an income distribution, inequality of incidence of temperature effects across different climates, inequality in the global distribution of income, and inequality in the distribution of warming.

Inference and persistence

The error bands depicted in Figure A3 reflect estimation imprecision corresponding to a significance level of $\alpha = 0.1$. This uncertainty is particularly important to include in this context since the volume of regressions underlying an LP-IRF precludes convenient summarization in a standard regression table. One may be additionally tempted to infer from the fact that the bands in red do not intersect the horizontal axis over the entire projection horizon that these estimates constitute sufficient evidence to reject a null hypothesis of level effects which persist for less than 15 years.

However, the uncertainty depicted by error bands correspond to individual hypothesis tests. Since estimates and residuals across an IRF are serially correlated—for example, the green bands straddle the horizontal axis but all estimates are negative— the relevant test of significance is a joint hypothesis test. Focusing on the impulse response corresponding to the reference temperature for convenience, the significance of the corresponding impulse response $\mathcal{R}_{\tau \to y}(h)$ amounts to testing the null hypothesis that all main effects are 0:

$$H_0(h) : \beta_{1,0} = \beta_{1,1} = ... = \beta_{1,h} = 0$$

Rejecting the null for at least one of the $h+1$ component main effects is sufficient to reject the null of the total impulse response up to horizon h. Since the

number of coefficients being considered in the joint hypothesis increases in h, the coverage of the joint hypothesis test must be adjusted for the inclusion of each additional horizon to compensate for the problem of multiple hypothesis testing. Inoue et al. (2024) derive convenient estimators for the implied "significance band" defined to satisfy the following condition:

$$\mathbb{P}\left[\bigcap_{h=0}^{H}\left\{\zeta_{\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}} < \widehat{\beta}_h < \zeta_{1-\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}}\right\}\right] \geq 1-\alpha$$

Note that the scaling of the significance level α by $2(H+1)$ is an implementation of the Bonferroni correction, admittedly considered overly conservative. Graphically, these bands are represented by a region straddling the horizontal axis such that estimates of β_{1h} contained within it are insufficiently large enough to reject the hypothesis of a null effect that the impulse response persists up to period h. The width of these bands increases with the projection horizon, corresponding to the fact that for each projection horizon, the available estimation sample is reduced by one period (in the case of a perfectly balanced panel).

These significance bands are specific to the main effect and are represented by the gray regions in the middle panels of Figure A3. Since ours is a state-dependent model, I derive equivalent significance bands specific to the interaction effects. These are defined relative to the point estimates of the main effects and are depicted by the gray regions in the panels on the left and right. Points outside these regions are considered sufficient to reject the null hypothesis that interaction effects do not persist up to h periods. By the linearity assumption, the null hypothesis of zero interaction effects for a 10° climate is rejected if and only if it is rejcted for a 26° climate.

For comparison, I include in Figure A4 a placebo test where I construct bands to implement significance tests equivalent to those described above to test the persistence in the effect of temperature on another widely available and potentially non-stationary growth rate: that of population. As I am not aware of any theory or evidence to suggest temperature shocks would meaningfully impact country-level population growth[6], we should expect the tests to return precise null results. Indeed, that is exactly what we find with all point estimates resulting from two separate population datasets easily contained by the corresponding significance bands. Notably, if we were to misuse

[6]While there is an established literature on weather impacts on mortality rates (Barreca and Shimshack, 2012; Barreca et al., 2016; Carleton et al., 2022), absolute magnitudes are miniscule relative to entire populations

the error bands as indicators of statistical significance, we would misinterpret the estimated IRFs as evidence to reject the null for large subintervals of the projection horizon for mid-temperature and hot climates. Instead, the entirety of these error bands are easily enveloped by the significance bands.

Since including significance bands, color-coded error bands, and point estimates for multiple IRFs can clutter space, I adopt a convention of omitting the significance bands entirely and instead coloring black the point estimates which are located outside the implicit error bands. In all IRFs beginning with those depicted in Figure 2, rounded points correspond to significant main effects while asterisks correspond to significant interaction effects. By this measure, these results cannot reject the null hypothesis of one-period level effects.

The state-dependent cumulative multiplier

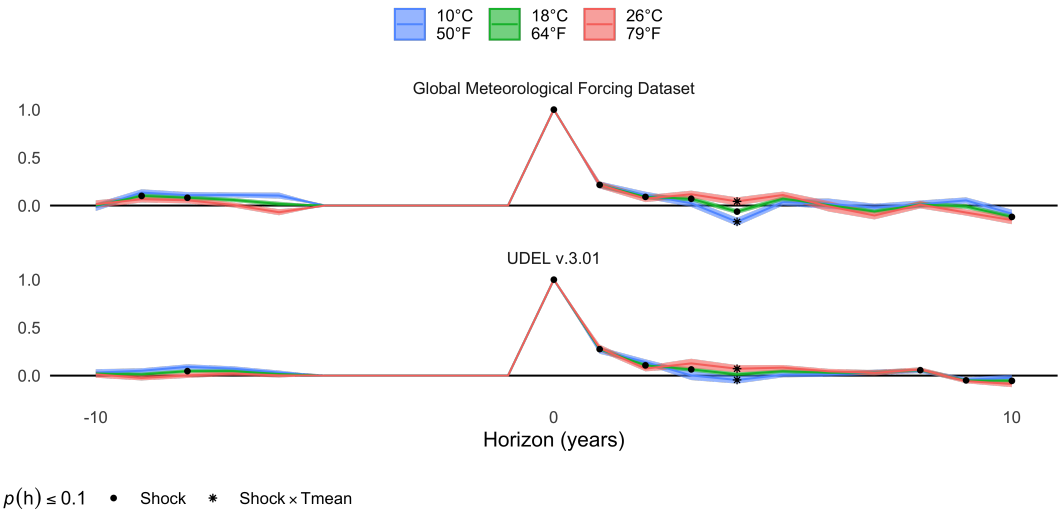
Nath et al. (2024) and Bilal and Känzig (2024) motivate their construction of temperature shocks by arguing that existing climate studies neglect to sufficiently account for the persistence of temperature in their estimated dynamic effects. An analogy is drawn to the custom of estimating multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be followed by additional stimulus programs in successive periods. Otherwise, the effect sizes implied by simple impulse responses functions misattributes cumulative changes in the outcome only to the initial unit shock; multipliers scale these effect sizes at each horizon by the corresponding accumulation of shocks.

Figure 1 depicts the impulse response of a unit temperature shock on itself for the three familiar representative climates. Each exhibits very similar persistence patterns across two independent temperature dataset. An application of the significance test from the previous subsection implies autocorrelations which persist for up to 10 years though only the first 1-2 periods are qualitatively large (approximately 21% and 9% respectively with insignificant heterogeneity across climates).[7] Cumulative multipliers are estimated in order to scale down the estimated impulse response of GDP $\mathcal{R}_{\tau\to y}(h)$ by the impulse response of the shock to itself $\mathcal{R}_{\tau\to\tau}(h)$ to account for this dynamic treatment schedule.

Cumulative responses are traditionally computed by first constructing *cumulative* versions of the two IRFs: $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to y}(j)$ for the outcome

[7]This contrasts with the finding of Nath et al. (2024), who measure shock persistence using levels of temperature and find that temperature levels remain between 0.1-0.2°C higher for at least nine years following a unit shock.

Figure 3: **Persistence of temperature shocks**



State-dependent impulse responses of identified temperature shocks estimated using two widely used temperature datasets each imply significant autocorrelation. In both cases, persistence is large in magnitude for the first 1-2 periods following a shock before flattening. Negative horizons are included as a check for pre-shock trends.

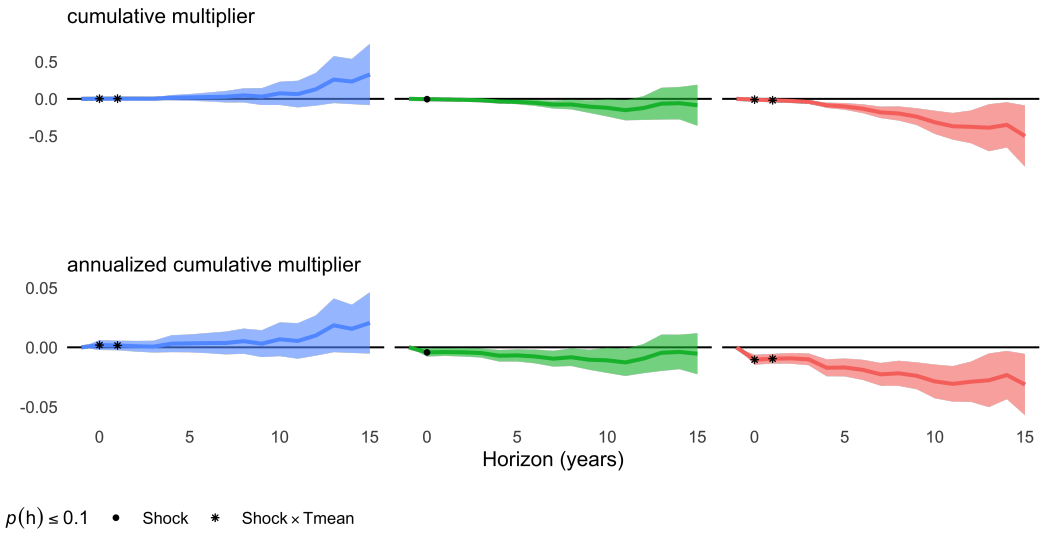
of interest y and $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to\tau}(j)$ for the shock $\widehat{\tau}$. Then the cumulative multiplier $m_{\tau\to y}(h)$ would be calculated as the ratio $\mathcal{R}^c_{\tau\to y}(h)/\mathcal{R}^c_{\tau\to\tau}(h)$. However, this approach has a couple of inconvenient downsides. First, as noted by Jordà (2024), computation of standard errors of a ratio of two random variables is complicated and non-standard. Second, it is computationally inefficient in the sense of requiring $2(h+1)$ ($4(h+1)$ in our state-dependent case) separate local projections to obtain the corresponding multiplier for each projection horizon h.

Instead, we follow the recommendation to use the one-step method described in Ramey (2016) and applied in Ramey and Zubairy (2018). We adapt the method for our state-dependent setting through the procedure summarized by the algorithm presented in in 1.

We can use the resulting coefficients to trace a cumulative multiplier function $\mathcal{R}^c_{\tau\to y}(h;\overline{T})$ analogous to (3) which accounts for the effect of persistence in the shock. Since estimation of the coefficients follows directly from local projection regressions, standard inference is preserved and the significance testing methods outlined in Section 2.2 are equally available. Figure 4 demonstrates an application of the method to estimate the "state-dependent temperature multiplier" of GDP. The estimated multiplier over horizon h is interpreted as the difference in cumulative GDP over the h periods following an isolated 1°C shock relative to a counterfactual where the shock had never occurred.

The second row provides the same estimates but annualized by dividing by $h+1$, interpreted as the average annual effect of the same isolated 1°C over the h periods following the shock. This latter formulation is helpful for interpreting outcomes which do not have an interpretation in cumulative terms. For example, the 'total impact on the the sum of Gini coefficients observed over h years' does not have economic meaning but the 'average annual effect of a 1°C shock on the Gini coefficient observed over h years' does.

Figure 4: **Cumulative GDP impulse response to a 1°C shock**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

3. Empirical results

We consider different outcomes Y related to within-country inequality and estimate state-dependent cumulative multipliers according to the following local projections for each projeciton horizon $h \in \{0, 1, ..., H\}$:

$$\Delta_h^c Y_{i,t+h} = m_{1,h}\widehat{\tau}^c_{i,t+h} + m_{2,h}\widehat{\tau}^c_{i,t+h}\overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h} \tag{4}$$

where $\mathbf{Z}_{it}$ include lag controls $\{\widehat{\tau}_{it}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta\text{GDP}_{i,t-j}, \Delta Y_{i,t-j}\}_{j=1}^{p}$

Figure 5: **Income share response to a 1°C shock**

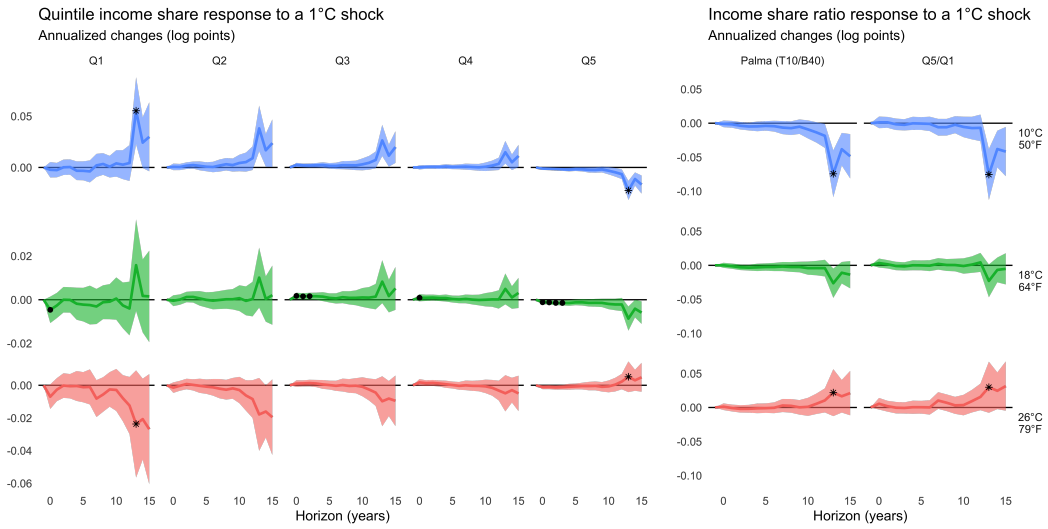
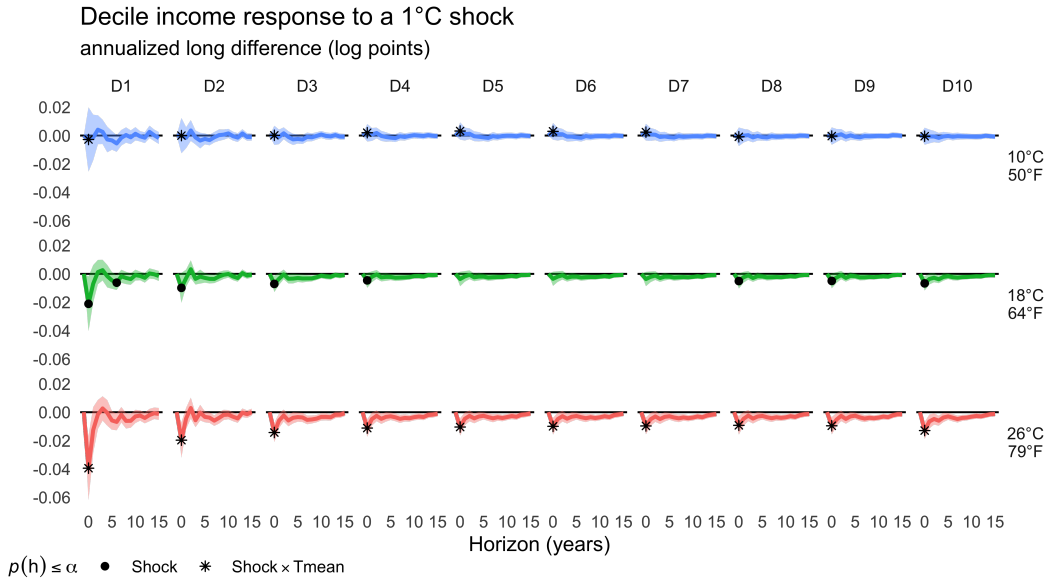


Figure 6: **Cumulative state-dependent decile income responses to a 1°C shock**



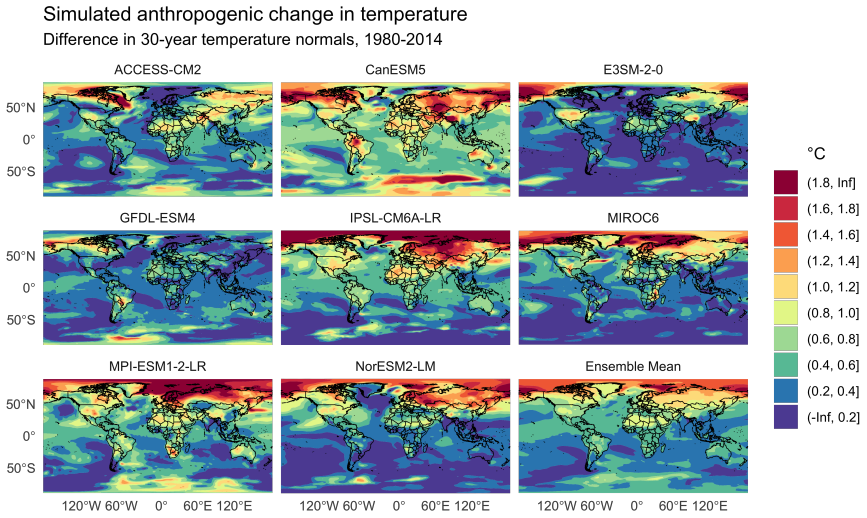
4. Welfare analysis

4.1 Implications for global inequality

4.2 Historical counterfactuals

Climate simulation data

Figure 7: **Model-implied spatial distributions of global warming, 1980-2014**



Climate simulation data come from the contributions of climate modeling groups to the World Climate Research Program's Coupled Model Intercomparison Project (CMIP). The CMIP, initiated in 1995, is a framework to coordinate, standardize, and disseminate the results of state-of-the-art simulations of the global climate for the benefit of the international climate research community. Data used in this study comes from CMIP6, the sixth and newest[8] generation of models (Eyring et al., 2016). Data is provided as a collection of globally gridded time series summarizing the joint evolution of several hundred climate variables under various pre-specified calibrations. We will primarily be interested in the models' simulations of near-surface air temperature.

[8] After a delayed rollout, outputs from most CMIP6 models were made accessible by 2022

All participating models are "coupled", meaning that they explicitly account for interactions and feedback between distinct components of the global climate system such as the atmosphere, the cryosphere, land surface, and the ocean. Because the manner in which this integration is achieved is idiosyncratic to each model, computationally expensive, and highly sensitive to external calibrations, coordinating the set of "experiments" that participating models run is necessary for comparability and interpretability. For example, one of the four indexes which define an experiment is the realization index which corresponds to the set of geophysical initial conditions at the beginning of the simulation period. By holding the initial conditions fixed, differences in results across models within the same experiment can be more readily attributed to the distinguishing features of the model without being confounded by differences in implementation. Similarly, differences in the same climate model's results across different perturbations to the initial conditions provide a measure of the model's internal variability. The CMIP thus provides a systematic way of decomposing total variation across models and experiments both internal and external to the model. Coordination also enables identification of systematic and idiosyncratic biases in model designs that then inform the next generation of models.

Among the other three indexes which define an experiment, the forcing index is central to our counterfactual analysis. Forcings are factors which affect the net transfer of radiative energy in the climate system, essentially the energy the Earth receives from the sun minus that which it expels back into outer space. Anthropogenic forcings include, for example, industrial greenhouse gas emissions, deforestation, and the transformation of land for arable agriculture. Natural forcings include aerosol expulsions from major volcanic activity and variation in solar irradiance such as those associated with the 11-year solar cycle.[9] Comparing simulations which only differ in their forcings allow one to attribute differences in outcomes to the differences in forcings; we describe next how comparisons of this kind form the basis of both our historical and projection counterfactual analyses.

Construction of historical counterfactuals

Our historical counterfactual analysis uses simulations from a category of experiments called "historical runs". For the CMIP6 generation of models, historical simulations begin at a pre-industrial baseline period in 1850 and run through to 2014. Among the forcing scenarios used in these historical experi-

[9]Natural forcings are distinct from sources of natural variability such as the ENSO cycle because they are external to coupled climate models.

ments is one simply labeled "historical", which attempts to represent the impacts of influential forcings actually imposed on the climate system over this period, such as the deforestation of the Amazon rainforest and the eruptions of Krakatoa in 1883 and Mt. Pinatubo in 1991. Results from this historical run are required of all participating CMIP6 models since these "hindcasts" can be directly compared to observational data in order to evaluate model accuracy. Models collectively perform remarkably well at reproducing the historical record (Zhou et al., 2020; Zelinka et al., 2020; Liu et al., 2021). For this reason, studies which use CMIP data for the purpose of counterfactual analysis are advised to include the full available "multi-model ensemble" for experiments of interest (Tebaldi and Knutti, 2007; Knutti et al., 2010; van Vuuren et al., 2006).

A subset of these models also simulate a set of optional 'historical-natural' experiments which are defined identically except with anthropogenic forcings held fixed at pre-industrial levels. The differences between the two forcings by the same model holding all other variables fixed is then interpreted as the model's simulation of anthropogenic contributions to climate change. Comparisons between historical and historical-natural experiments are commonly used in the subfield of attribution science concerned with quantitatively measuring anthropogenic contributions to the intensity, frequency, or probability of weather events or trends.

For each climate model m and country i, we use an observed series T and the two model-specific historical series $\mathcal{T}^{hist}$ and $\mathcal{T}^{nat}$ to construct a counterfactual series $\widetilde{T}$ where anthropogenic forcings have been held fixed since the first year of the simulation period. The iterative process is summarized in Algorithm 2 and illustrated in Figure 8 using data from the Philippines as an example. In panel (a), differences between model-specific simulations of national temperature when including all forcings (orange) and when excluding anthropogenic forcings (green horizontal axis). Local regression fits are depicted in red. Light-blue shaded regions correspond to 30-year reference periods used to define temperature normals (dark blue dots) for the start and end of the simulation period 1980-2014.

In panel (b), The linearized difference between normals depicted as dark blue lines in (a) are subtracted from the observed temperature series (black) starting in 1980 to produce counterfactual temperature series (green). Each of these is interpreted as the temperature history that would have been realized had anthropogenic forcings been held fixed at 1980 levels according to a specific model of the climate system.

In panel (c), linearizing the difference between the climate normals defined at either end of the simulation period and subtracting the result from

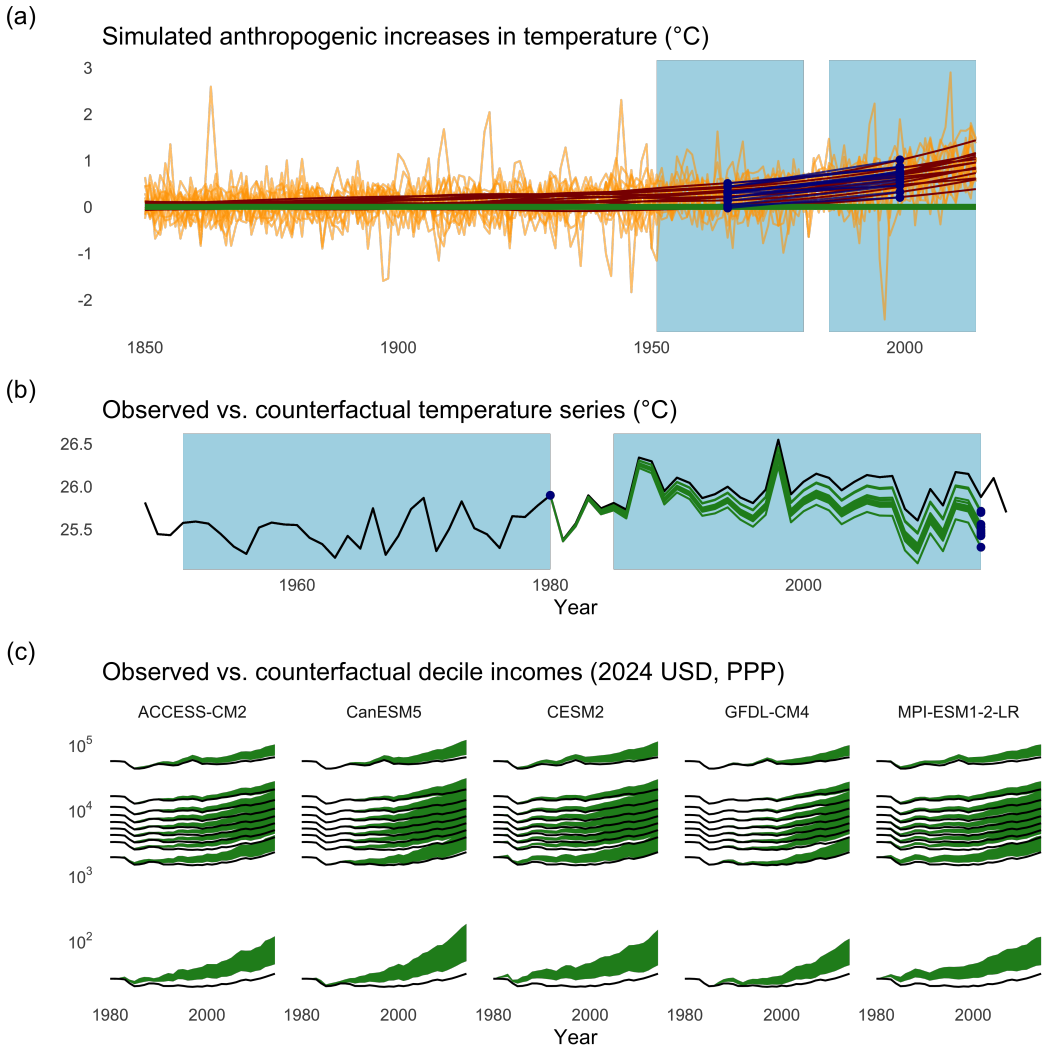
an observed temperature series implements the standard "delta method" for constructing bias-corrected climate counterfactuals. Counterfactual economic series for a random selection of five climate models are depicted in green and are constructed by iteratively applying the decile-level dynamic income responses depicted in Figure 6 to the observed and counterfactual temperatures depicted in (b). Their difference is interpreted as the change in income attributable to anthropogenic activity since 1980. Green bands correspond to 90% confidence intervals generated from 500 bootstrap estimates of the response functions.

4.3 Counterfactual results

Anthropogenic contributions to observed inequality since 1980

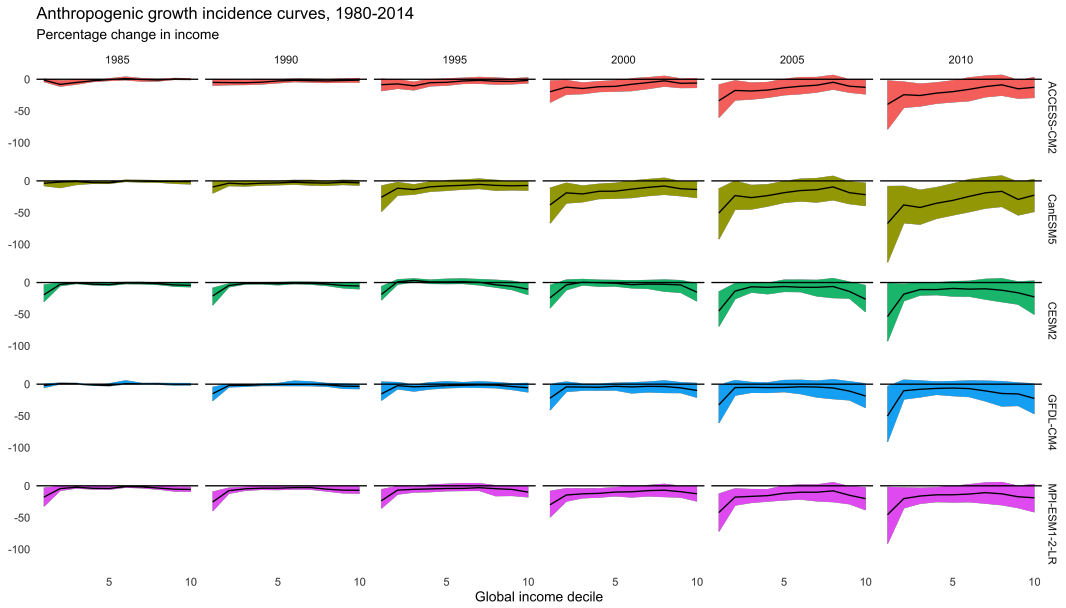
Global inequality decomposition: within-country vs. between-country

Figure 8: **Constructing historical counterfactuals holding anthropogenic forcings fixed at 1980 levels**



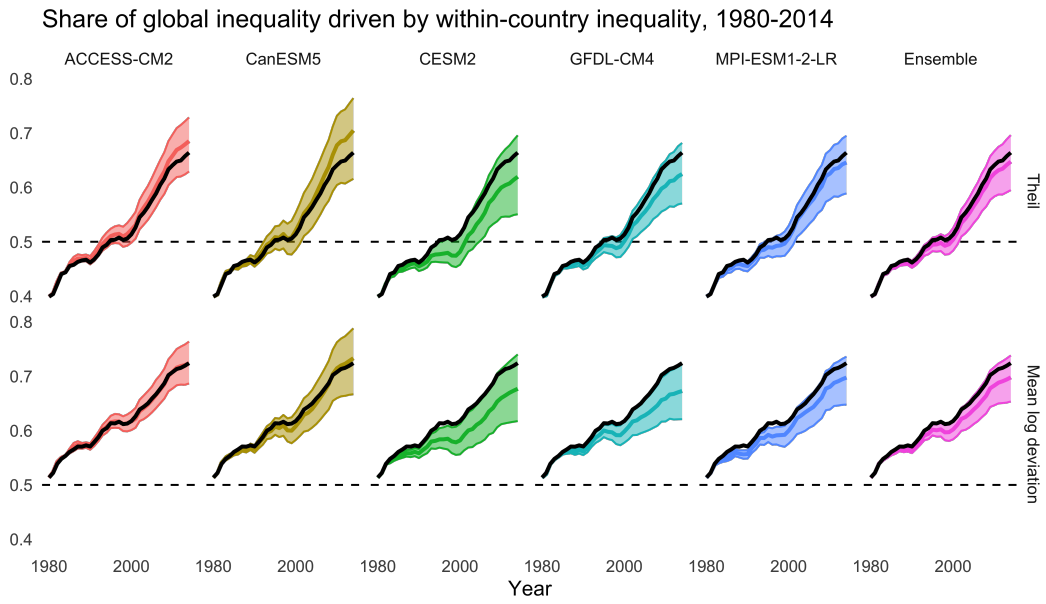
Demonstration of the delta method for constructing bias-corrected counterfactuals using data for the Philippines as an example.

Figure 9: **Anthropogenically attributable change in income by global income percentile**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

Figure 10: **Within-country and between-country inequality, observed and counterfactual**



5. Discussion

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5.1 Conclusion

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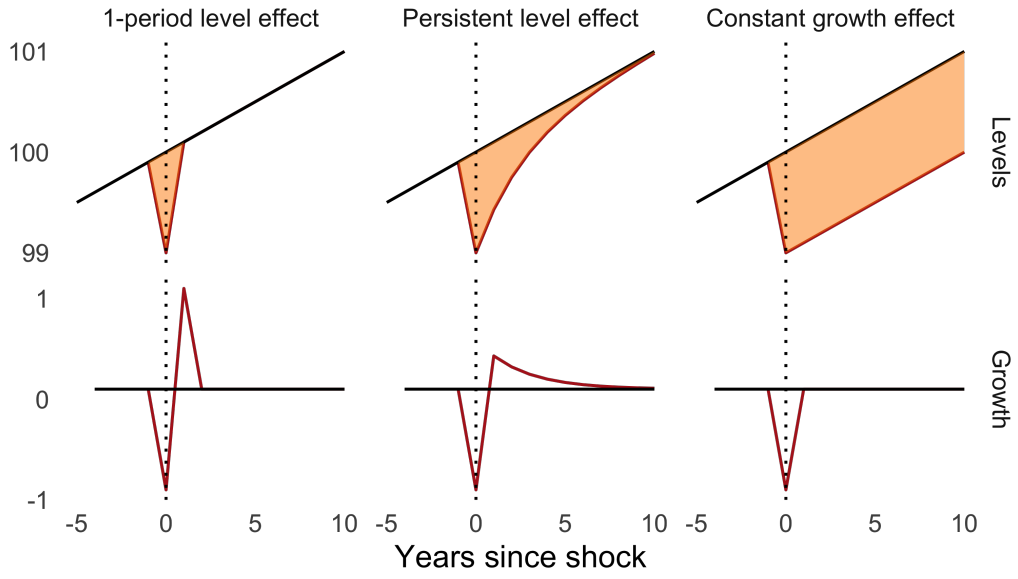
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A1. Data and methods

A1.1 Estimating persistence: growth vs. level effects

Figure A1: **Influence of persistence on levels and growth**



A1.2 Algorithms

Algorithm 1: One-step state-dependent cumulative multiplier

for $h = 0$ **to** H **do**

Estimate the IRF $\mathcal{R}_{\tau\to\tau}(h)$ as defined in (3), collecting state-dependent shock coefficients $\widehat{\alpha}_{1,h}, \widehat{\alpha}_{2,h}$;

Define $\Delta_h^c y_{i,t+h} := \sum_{j=0}^{h} \Delta_j y_{i,t+j}$;

Define $\widehat{\tau}^c_{i,t+h} := \widehat{\tau}_{it}\left[\sum_{j=0}^{h} \widehat{\alpha}_{1,j} + \widehat{\alpha}_{2,j}\overline{T}_{it}\right]$;

Estimate local projection:

$\Delta_h^c y_{i,t+h} = m_{1,h}\widehat{\tau}^c_{i,t+h} + m_{2,h}\widehat{\tau}^c_{i,t+h}\overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h}$;

end

Output: Multiplier coefficients $\widehat{m}_{1,h}, \widehat{m}_{2,h}, \widehat{\lambda}_h$ for $h \in \{0, ...H\}$

Algorithm 2: Constructing temperature counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\mathcal{T}^{hist}_{m,i,t}\}$,$\{\mathcal{T}^{nat}_{m,i,t}\}$,$\{T_{it}\}$

for $t^* \in \{1980, 2014\}$ **do**

$\overline{\mathcal{T}}^{hist}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{hist}_{m,i,t^*-j}$;

$\overline{\mathcal{T}}^{nat}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{nat}_{m,i,t^*-j}$;

$\overline{\delta}_{m,i,t^*} \leftarrow \overline{\mathcal{T}}^{hist}_{m,i,t^*} - \overline{\mathcal{T}}^{nat}_{m,i,t^*}$;

end

for $t = 1980$ **to** 2014 **do**

$\overline{\delta}_{m,i,t} \leftarrow (t - t_0)\frac{\overline{\delta}_{m,i,2014} - \overline{\delta}_{m,i,1980}}{2014-1980}$;

$\widetilde{T}^{nat}_{m,i,t} \leftarrow T_{it} - \overline{\delta}_{m,i,t}$;

end

Construct states $\{\overline{T}^{nat}_{m,i,t}\}$ by applying (1a) to $\{\widetilde{T}^{nat}_{m,i,t}\}$ then subtracting by $\widetilde{T}^{nat}_{m,i,1980}$;

Construct shocks $\{\widehat{\tau}^{nat}_{m,i,t}\}$ by applying (1a) and (1b) to $\{\mathcal{T}^{nat}_{m,i,t}\}$;

Output: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$

Algorithm 3: Constructing economic counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$, B bootstrap samples indexed b

Initializing series:

  for $b = 1$ **to** B **do**

    Specify desired simulation interval: $(t_0, t_1) \leftarrow (1980, 2014)$;

    Estimate $\mathcal{R}^{c}_{\tau \to Y}(h; \overline{T}, b)$ from (4);

    Specify maximum projection horizon H using persistence test from Section 2.2;

    Define IRFs $\mathcal{R}_{\tau \to Y}(h; \overline{T}, b) := \mathcal{R}^{c}_{\tau \to Y}(h; \overline{T}, b) - \mathcal{R}^{c}_{\tau \to Y}(h-1; \overline{T}, b)$;

    Collect IRF coefficients;

    $\{\widehat{\beta}_{1,h,b}, \widehat{\beta}_{2,h,b}, \widehat{\lambda}_{h,b}\}_{h=0}^{H}$;

    Define $\delta^{\tau}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} - \widehat{\tau}_{it}$;

    Define $\delta^{\tau T}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} \cdot \overline{T}^{nat} - \widehat{\tau}_{it} \cdot \overline{T}_{it}$;

    Define $\delta^{T}_{m,b,i,t} := \overline{T}^{nat}_{m,i,t} - \overline{T}_{it}$;

    Initialize $\delta^{Y}_{m,b,i,t} \leftarrow 0$;

    Initialize $\widetilde{Y}^{nat}_{m,b,i,t_0} := Y_{i,t_0}$

  end

Iteratively constructing series:

  for $t = t_0 + 1$ **to** t_1 **do**

    for $b = 1$ **to** B **do**

      for $h = 0$ **to** H **do**

        $\delta^{Y}_{m,b,i,t+h} \leftarrow \delta^{Y}_{m,b,i,t+h} + \widehat{\beta}_{1,h,b} \cdot \delta^{\tau}_{m,b,i,t} + \widehat{\beta}_{2,h,b} \cdot \delta^{\tau T}_{m,b,i,t} + \widehat{\lambda}_{h,b} \cdot \delta^{T}_{m,b,i,t}$;

      end

      $\widetilde{Y}^{nat}_{m,b,i,t} \leftarrow \left(1 + \log(\frac{Y_{it}}{Y_{i,t-1}}) + \delta^{Y}_{m,b,i,t}\right) \widetilde{Y}^{nat}_{m,b,i,t-1}$

    end

  end

Output: $\widetilde{N}^{nat}_{m,b,i,t}$

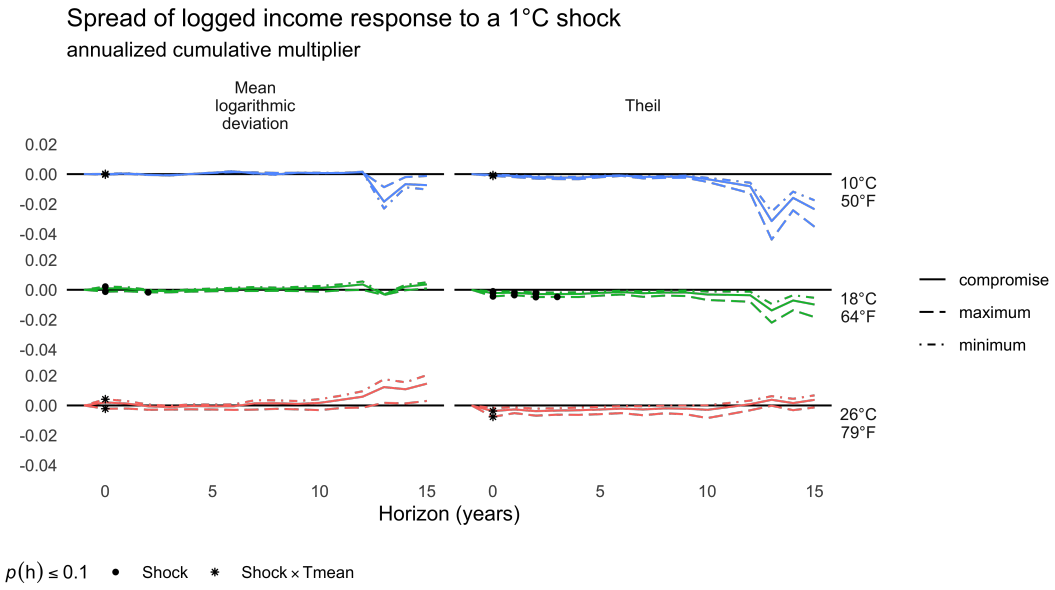
A1.3 Construction of inequality indices

Inequality data from the World Inequality Database is provided at the country-percentile level with variables reporting averages, minimums, maximums, and shares of national income. As grouped data, its use for the construction of inequality indices still requires the modeler to impose assumptions about the distribution of household incomes within these groups. This becomes partic-

ularly influential for the construction of entropy-based inequality measures such as the mean logarithmic deviation or the Theil index.

For a given inequality measure J, the lower bound J_L is calculated by assuming perfect equality within all groups such that assigning all individuals their group mean income. The upper bound J_U is calculated by assigning a proportion of the individuals within a group the minimum income of the group and the remainder the maximum income of the group[10].

Figure A2: **Sensitivity of inequality index effects to assumptions about within-percentile inequality**



Cowell (2011) describes multiple 'compromise' assumptions between the two extremes, finding that they each approximate an average where the lower bound is given twice the weight of the upper bound. Figure A2 depicts IRFs of the two inequality measures under each of these three assumptions.

A2. Supplementary empirical results

[10]We omit the formula but within-group inequality is not generally maximized by an equal split between the two

Figure A3: **GDP impulse response to a 1°C shock, including significance bands**

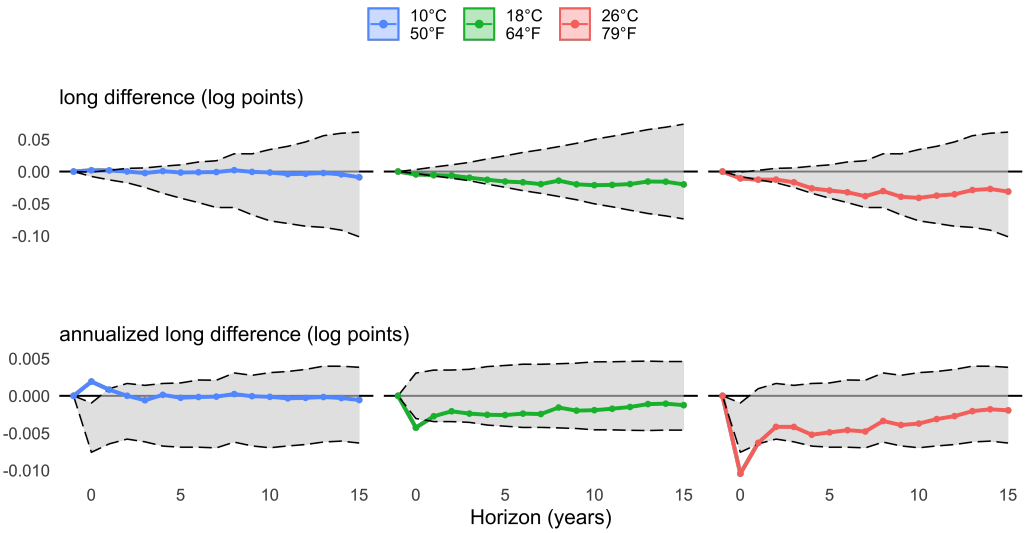


Figure A4: **Estimating national population impulse response to a 1°C shock as a placebo test**

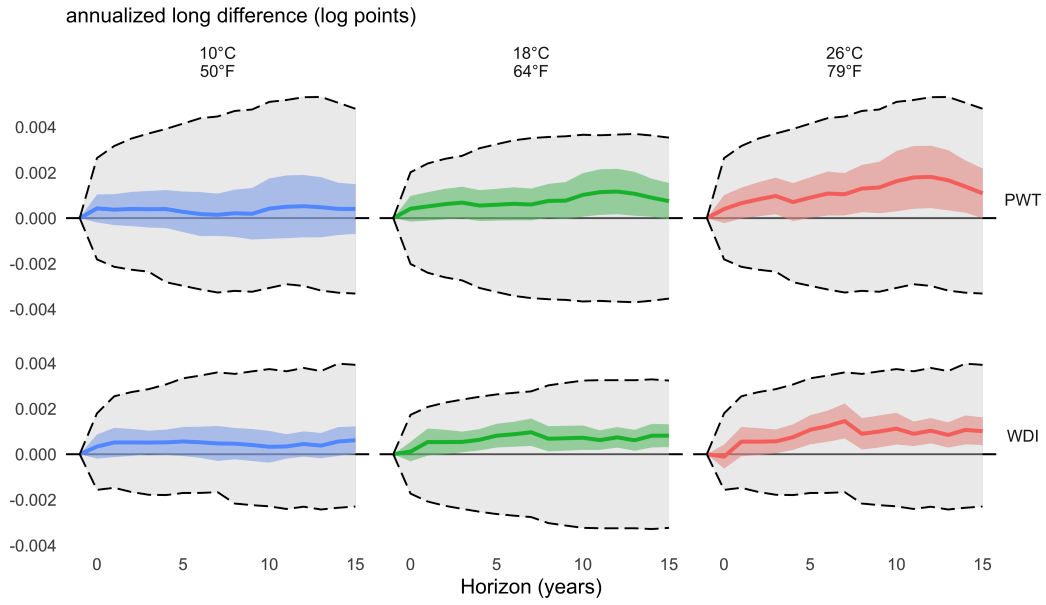


Figure A5: **Inequality index response to a 1°C shock**

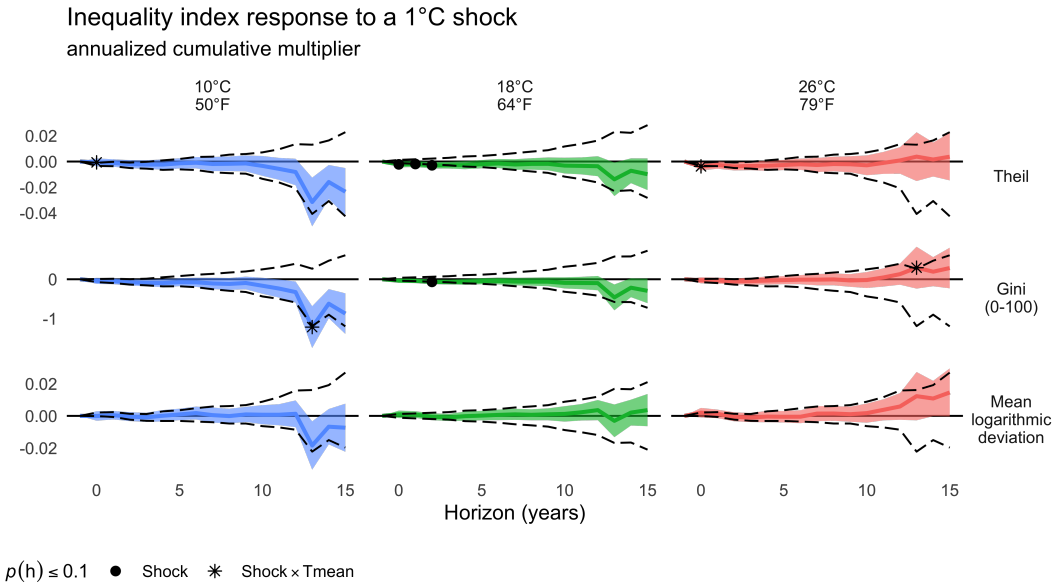
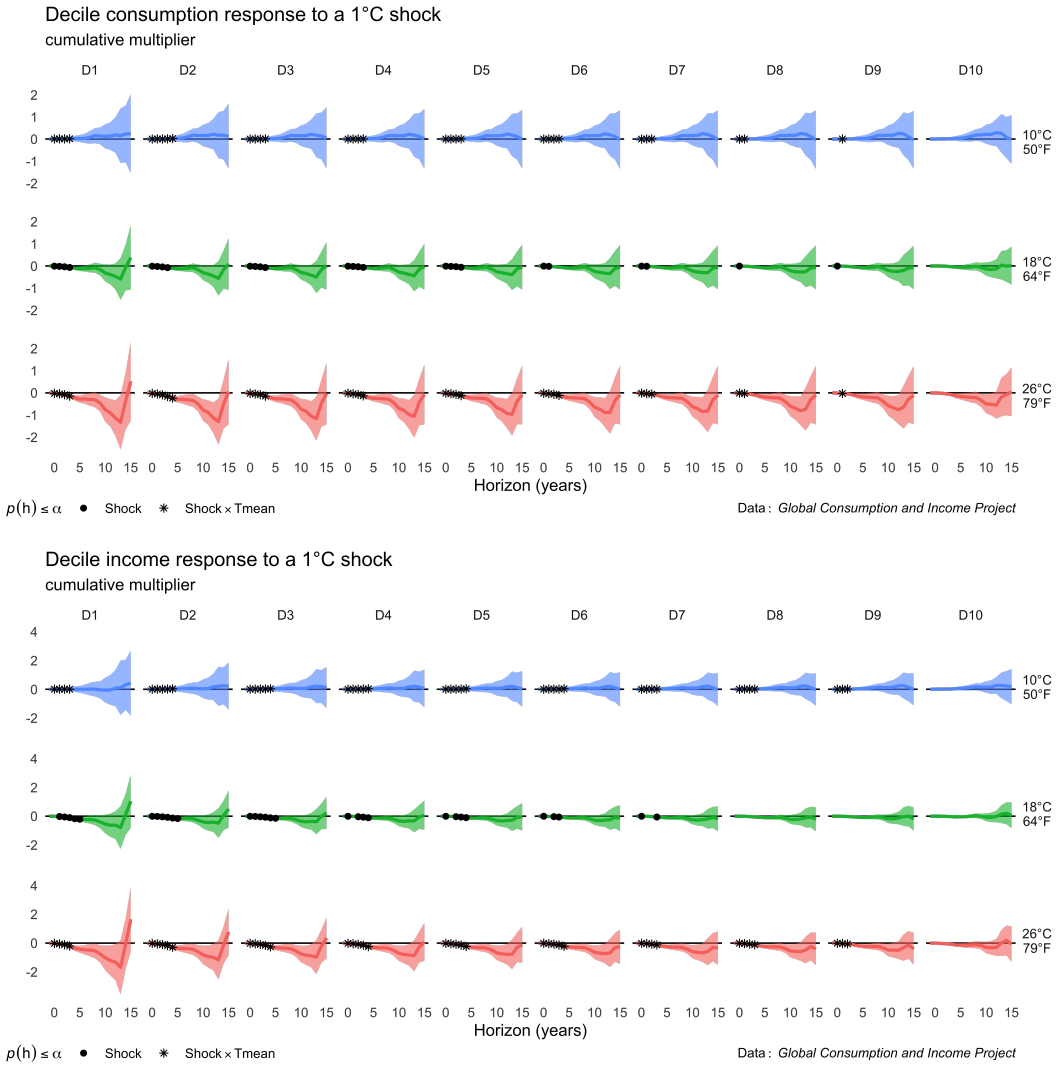
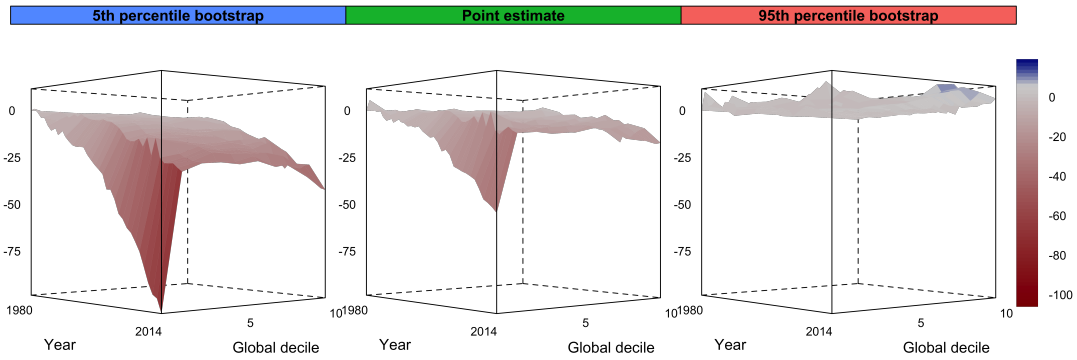


Figure A6: **Distribution of consumption and income effects measured by survey microdata**



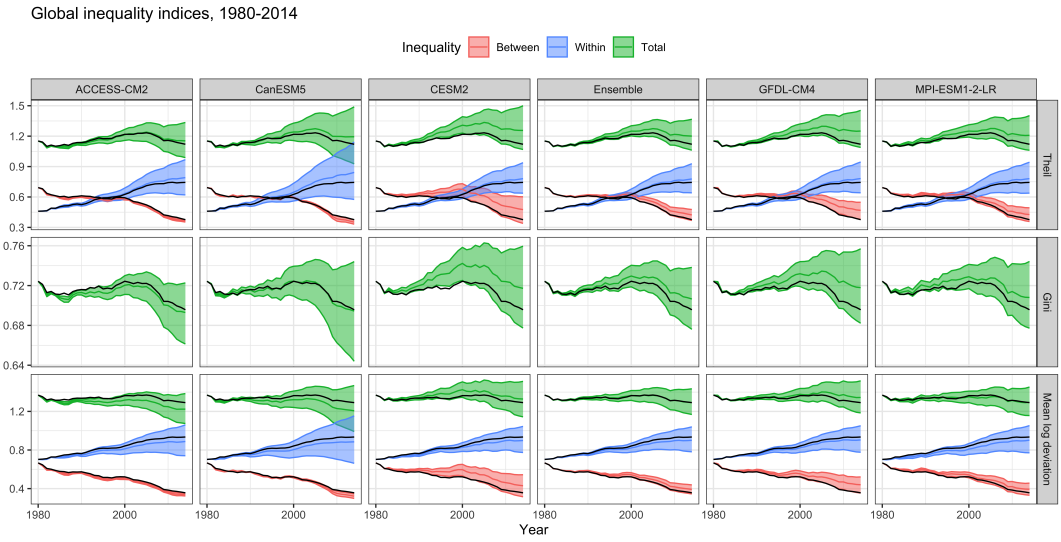
A3. Supplementary simulation results

Figure A7: **Anthropogenically attributable change in income by global income percentile**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.

Figure A8: **Measures of global inequality, observed and counterfactual**



The cumulative multiplier $m(h)$ is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future values of $\widehat{\tau}$.